⚡ Charges & Static: Guided Notes · ANSWER KEY

Chemistry · Unit 1A · Level 3 (closer to the class work) · goes with L1.1-L1.6 (Initial Model, Smaller Scales, Sparks, Particle Model of Static, Cause of Static)

Name: _____

Use the class slides and your reading. Key words:

atom · *nucleus* · *proton* · *neutron* · *electron* · *charge* · *neutral* · *static charge* · *correlation* · *causation*

A. Lightning and evidence (L1.1, L1.2)

1. Lightning is a giant electric **spark** between a cloud and the ground.
2. Rain and lightning happen together. This is a **correlation**.
3. Why does a correlation NOT prove that rain causes lightning?
Two things can happen together because of a third cause; we need evidence of the mechanism.

B. The atom (L1.5)

4. Fill in the table.

Particle	Charge	Where is it?	Can it move to another object?
Proton	**+ (positive)**	**nucleus**	**no**
Neutron	**0 (none)**	**nucleus**	**no**
Electron	**− (negative)**	**outside the nucleus**	**yes**

5. An atom with equal numbers of protons and electrons is **neutral**.
6. Draw a neutral atom with 3 protons, 3 neutrons and 3 electrons. Label each particle and its charge.
3 + in the nucleus with 3 neutral neutrons, 3 − around the outside, all labeled.

C. Static charge and sparks (L1.3, L1.6)

7. Rubbing a balloon on hair moves **electrons** to the balloon.
8. The balloon becomes **negative** and the hair becomes **positive**.
9. Explain why the hair is now positive. Use the words electron and proton.
The hair lost electrons but kept all its protons, so it has more + than −.
10. What is a spark? Use the word charge.
Built-up charge (electrons) jumping across a gap of air.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*